patent application does not contain detailed information of how to put the invention into practice, this requires less substantial arguments and evidence. Serious doubts whether the skilled person can carry out the invention as claimed, e.g. in the form of comprehensible and **plausible** arguments, are sufficient (see also **T 347/15**; **T 1845/17** (ambiguous parameter); **T 2218/16** (Gene therapy of motor neuron disorders – scAAV9 vector), which summarises the case law on the burden of proving an alleged lack of sufficient disclosure).

The board in **T 59/18** stated, in relation to the burden of proof when a weak presumption exists that the invention is sufficiently disclosed (reference made to **T 63/06**), that since the respondents (opponents) had convincingly argued that the term "relaxation ratio" having the meaning intended by the appellant was not to be found in textbooks and did not represent common general knowledge, the burden of proof for establishing sufficiency was clearly on the appellant/patent proprietor.

In **T 2119/14** (unusual parameter) the board, referring to **T 63/06** (concepts of strong presumption and weak presumption in relation to the burden of proof), held that the existence of an undue burden resulted from the almost infinite number of coating compositions that fell under the structural definition given in claim 1, and the established absence of a teaching in the patent in suit as to how select in an appropriate and straightforward manner the components of the coating composition so as to meet the unusual parametric requirement of claim 1. As a consequence, the onus of proving that the preparation of the coating compositions over the whole scope for which protection was sought did not necessitate an undue amount of work for the skilled person rested with the patent proprietor (here: appellant).

9.2. Objection of insufficient disclosure rejected – examples

The established case law of the boards of appeal is that a finding of lack of sufficient disclosure should be based on serious doubts, substantiated by verifiable facts. The facts put forward by the examining division in ex parte case **T 1020/11** to justify a finding of lack of sufficient disclosure were based on a potential problem which might occur between different antigens in combination. There were however no verifiable facts on file that demonstrated that interference was a problem in the present, specific case. In the absence of such verifiable facts relevant to the specific case, the board could not find the objection of lack of sufficient disclosure persuasive (see also inter partes case **T 872/13** concerning a pharmaceutical composition) – the opponent described a number of possible difficulties that the skilled person might encounter, but did not raise any serious doubts substantiated by verifiable facts – in this case the skilled person was in a position to modify the method of Example 8 in order to obtain the desired result – routine modifications).

T 1437/07 (Botulinum toxin for treating smooth muscle spasm) dealt with the objection that it was not credible that the therapeutic effect could be achieved because the treatment disclosed in Example 9 had not actually been carried out. The board referred inter alia to R. 42(1)(e) EPC, according to which not even the presence of an example was mandatory. Therefore, just because a patent disclosed an effect which had not in reality been achieved, there was no reason – in the absence of convincing evidence that the effect could not be achieved – for the board to doubt that the effect could be achieved. The board rejected the objection.

As proof that an invention has been insufficiently disclosed, the boards require that the attempt to repeat it must fail despite following the conditions given in the examples. This requirement is not fulfilled where the opponent repeats the patented process under conditions covered by claim 1 but differing in many respects from those applying in the contested patent's examples (**T 665/90**).

An invention should be reproduced using the examples given. Insufficiency cannot be proven on the basis of laboratory trials when the only embodiment exemplified in the patent is an industrial fermentation process (**T 740/90**). The disclosure was also considered sufficient where the opponent had only used equivalents of the surfactants given in the patent, as he had not discharged his burden of proof (**T 406/91**).

In **T 1712/09**, the board held that the opponent had failed to prove that the method of measuring the parameters was unworkable. The tests referred to in its experimental reports had been carried out using measuring apparatus different from that described in the patent, and so not as instructed there. The board found that no attempt had been made to reproduce the invention (no attempt at calibration), which was the first condition for an objection under Art. 100(b) and 83 EPC. It cited **T 815/07** (need for consistent values) and **T 1062/98** and **T 485/00** (possibility of calibrating methods of determining the relevant parameters). The board in **T 548/13** held that the case law on parameters (which included **T 815/07**) did not apply since the case before it did not concern a quantitative parameter.

In **T 45/09** too, the opponent's test conditions were called into question since the tests had been carried out using a commercially available product. Observing that two products of the same brand but available on the market at different times would not necessarily have the same properties, the board found that it had not been established that the properties had been the same in this specific case. The board also considered the issue of calibrating the method of measuring the parameter. The board concluded that the opponent had failed to show that the method of measuring the parameter could not be reproduced and thus to prove insufficient disclosure. It was, indeed, for the opponent to do so, and it could have discharged its burden of proof by attempting to reproduce the method using at least one of the claimed silica.

In **T 378/11** (particle size), the values disclosed in claim 1 referred to the average particle size. **No proof** had been submitted by the appellants/opponents that the lack of any indication as to the precise kind of average particle size hindered the skilled person from carrying out the invention at issue. The board found that the skilled person could select those kinds of mean values falling within the range 10-500 mym in order to prepare a composition according to the invention at issue. Whether or not the use of two kinds of mean values would lead to different results was a matter of clarity, rather than sufficiency of disclosure.

In **T 1089/15** the board, recalling the established case law on a successful objection (e.g. **T 19/90** OJ 1990, 476, point 3.3 of the Reasons), found that the appellant (opponent) had not filed any evidence in support of its case and relied in its arguments on mere allegation. At the oral proceedings too, the appellant had merely referred to its written submissions, and thereby chosen not to submit arguments countering the position expressed by the board in the communication (Art. 15(1) RPBA) sent in preparation for those oral proceedings.

9.3. Burden of proof on the opponent – special cases

In **T 347/15** the board endorsed the case law on evidence (Art. 83 EPC) and, having found that the application did not contain detailed information about how to put the invention into practice, applied the reasoning given in **T 491/08**. In the case in hand, the opponent (appellant II) would have had to perform a large number of costly tests to prove that the invention could not be carried out, whereas a single example from appellant I (patent proprietor) would have been enough to prove that it could. The board concluded that appellant II (opponent) had discharged its burden of proof by submitting comprehensible and plausible arguments giving rise to serious doubts about whether the skilled person could carry out the invention as claimed. That meant it had then been up to appellant I (proprietor) to prove the contrary, i.e. that the invention could be carried out, but it had failed to do so. In particular, the board found that, owing to the lack of information in the patent, carrying out the invention went beyond the usual know-how of the skilled person.

In **T 970/16** (use of the cellulase in laundering process), the board stated that the opponents, which indeed did not file themselves any evidence, discharged their burden of proof by relying on the available evidence X29 (experimental report filed by the patent proprietor) and by plausibly arguing that common general knowledge would not enable the skilled person to realize the disputed benefits. The required technical effect was due in the present case to a hypothetical mechanism not yet being part of common general knowledge as stated in the patent. The skilled person, in view of the unsuccessful results of X29, would be obliged to start a research program.

9.4. Taking account of doubts

An appellant (opponent) is not required to prove that carrying out the invention is inherently impossible; it need only adduce understandable arguments casting doubt on whether the invention can be carried out over the whole scope of the conferred protection on the basis of the patent specification and, if necessary, common general knowledge (**T 809/13**).

The board in **T 1299/15** observed that it was established case law that the opponent (here: appellant) bore the burden of proving an alleged lack of enablement. However, since the patent specification did not describe any embodiment, the appellant in this case could not take such an embodiment as a basis for proving the lack of enablement; instead, it could only invoke a lack of **plausibility** in support of its case. Having done precisely that, it had shifted the burden of proving the contrary to the patent proprietor (respondent) and, that being the case, the board ultimately decided that the **doubts** had to go that party's detriment. (See also **T 805/17** on the burden of proof and means of evidence in a case where a patent contained so little information that it was considered insufficient even in combination with the skilled person's common general knowledge – dental reconstruction – doubts about availability of a method of manufacturing a blank with the claimed properties).

In ex parte case **T 2249/16** the board applied the principle that doubts need not be proven by way of experiments but must at least be substantiated in a technically plausible manner. It observed that the contested decision did not set out any reasons or arguments as to why the skilled person would be unable to carry out the invention.

9.5. Burden of proof wrongly allocated at first instance

T 55/18 is an example of a case where the board found that the opposition division had been wrong to put the burden of proof on the patent proprietor. On this shifting of the burden of proof, the board held, in particular, that the fact that the appellant (patent proprietor) had filed two pixelated and fuzzy photos (obtained by optical microscopy – means not appropriate for investigating the presence of nanometer particles in suspension) could not as such relieve the respondent (opponent) of the burden of proof. Ultimately, the objection was found not to be substantiated by verifiable facts. Since the respondent had not shown that a skilled person could not carry out the invention, or at least not convincingly put the ability to do so in **doubt**, a lack of sufficient disclosure could not be established. One of the respondent's arguments was that every step of the claimed method had to be sufficiently disclosed, but the patent did not disclose how to carry out step (d) because there was no specific indication of the dimension or the method for measuring it. But, observing that claim 1 did not comprise a step of measuring the dimension of the suspension elements, the board held that the absence of this specific indication did not affect the reproducibility.

In **T 2571/12** the board disagreed with the conclusions of the opposition division that because no evidence had been provided by the opponent to show that any neuropsychiatric disorder could not effectively be treated using a glutathione precursor the patent in suit was considered as disclosing the invention in a manner sufficiently clear and complete for it to be carried out by a person skilled in the art. It is the patent that has to demonstrate the suitability of the claimed treatment for the claimed therapeutic indication. As explained, for example, in decision **T 609/02**, a simple verbal statement that compound X may be used to treat disease Y was not enough to ensure sufficiency of disclosure: rather, it is required that the patent provides some information in the form of, for example, experimental tests, to the avail that the claimed compound has a direct effect on a metabolic mechanism specifically involved in the disease, this mechanism being either known from the prior art or demonstrated in the patent per se.

9.6. Consequence of an invention going against generally accepted scientific principles

It was held in **T 541/96** that if an invention seems to offend against the generally accepted laws of physics and established theories, the disclosure should be detailed enough to prove to a skilled person conversant with mainstream science and technology that the invention is indeed feasible, the onus being on the applicant (see also **T 1023/00**, **T 1329/07**, **T 1796/07**, **T 1603/13** (violation of laws of physics – thermodynamics – energy conservation), T 1485/17 (low frequency signal inherent to only biological systems)). The more a new invention contradicts previously accepted technical wisdom, the greater the amount of technical information and explanation is required in the application to enable the invention to be carried out by the average skilled person to whom only that conventional knowledge is available (T 1785/06).

In ex parte case T 2340/12, the application related to a space energy implosion unit. The board observed that it did not understand how the torsion field or space energy was to be measured. The appellant (applicant) claimed that over 40,000 Internet citations could be found concerning "Space Energy". But no specific Internet citation was cited which could serve to explain the concepts of torsion field or space energy. The applicant only referred to "indirect" measurements carried out but did not elaborate on the nature of these experiments or on their relevance for the claimed invention despite having been invited to do so in the provisional opinion issued by the board. The examining division raised criticisms regarding the experiments. The appellant emphasised that the EPC does not contain any requirements for such experimental evidence to be provided. The appellant further questioned the competence of the examining division to require such evidence. The board stated that in the case of inventions in **fields of technology without any accepted theoretical or practical basis**, the case law of the boards of appeal has established that the application should contain all the details of the invention required for the effect to be achieved (cf. T 541/96, point 6.2 of the Reasons). This was the direct consequence of the fact that the skilled person will be unable to rely on common and accepted general knowledge when dealing with inventions in such fields. The board stated that there is no provision in the EPC according to which the grant of a patent depends on the filing by the applicant of evidence that the claimed invention performs satisfactorily in the form of results of experimentation. The filing of such results is **not** to be seen as **an obligation** imposed on the applicant **but**, in contrast, as a right, providing the applicant with the **opportunity to convince** the examining division (or the board) that it erred in its initial findings. The decision includes findings on the burden of proof in **ex parte** cases (see for example, the summary of the rules established by the case law for inter partes cases given in T 967/09, point 6 of the Reasons).

In T 518/10, the board looked at the rules concerning the burden of proof of insufficiency, which was, as a general rule, on the opponents. In the case at issue, the appellant (patent proprietor) had asserted that, against the prevailing technical opinion, by using the extraction method described in the patent in suit the skilled person was able to obtain from marine and aquatic animal material an extract comprising compound (II). The respondents denied this and provided evidence that compound (II) could not be obtained when working according to the general method described in the patent. Under these circumstances the burden of proof was on the appellant to show that the method in the patent worked as alleged. The mere assumption that compound (II) could theoretically be present in an extract due to the krill's diet on algae was not evidence that could disprove the respondents' experimental reports or discharge the burden of proof resting on the appellant. The board also did not share the appellant's opinion that it was for the respondents, after having failed to obtain the claimed extract by following the teaching of the patent, to embark on a **research programme** in an attempt to find a compound which, according to the prevailing technical opinion, was not expected to be found in the first